

AgriDrone Network

A DePIN marketplace on Solana for agricultural drone spraying — landscape, critique, and a five-week build plan for a hackathon submission.

The idea is real and the market is real. The distributor relationship is a genuinely rare asset that is currently buried under a token model that weakens the pitch. Three changes turn this from “another drone escrow project that didn’t place” into a serious submission.

PREPARED FOR

Khristo, Tbilisi

DATE OF ANALYSIS

21 September 2026

EVIDENCE BASE

5,400+ builder projects · 6,300+ Grid products · crypto archives · web

The verdict in one page

Three changes that decide this submission

- 1. **Drop the token and say why.** Your design is peer-to-peer, and a token subsidy in a peer-to-peer network is a self-dealing attack surface. Settle in USDC, bond the operator in USDC, and make the deliberate absence of a token part of the pitch.
- 2. **Verify dispensed volume, not flight path.** A GPS track is identical whether the pump is on or off. This is a proven impossibility, not an engineering gap. Flow-meter volume over area covered is the signal that separates spraying from flying.
- 3. **Get one real job sprayed and settled before the deadline.** One filmed field beats every feature you could build instead.

The regulatory question is the one thing that could invalidate the premise outright, and it is currently a hole in the plan. Georgia’s Open category explicitly excludes any operation that drops material, which puts agricultural spraying categorically outside it. Email UAS-GA@gcaa.ge this week.

How this is likely to score

Real problem, real users	● Strong	Physical, specific, non-crypto-native users
Founder-market fit	● Strong	The DJI distributor relationship is rare
Working prototype	▲ At risk	New to Anchor — scope ruthlessly
Solana integration clarity	▲ At risk	A named common failure mode
Novelty	◆ Medium	DroneForce did the generic version in 2025
Venture scale	◆ Medium	Georgia alone is not venture scale
Team	■ Weak if solo	Average winning team size is above three

Figure 1. Projected performance against Colosseum’s published judging emphasis. Derived from “Perfecting Your Hackathon Submission” (May 2025) and “How to Win a Colosseum Hackathon” (Feb 2024).

Track fit

Go in under **DePIN** as the primary track. In Frontier it carried 84 projects against 493 in AI Platforms and 438 in Consumer Apps — one of the least crowded tracks that still has a dedicated prize. **Marketplace Platforms** (50) or **RWA** (98) are reasonable secondaries.

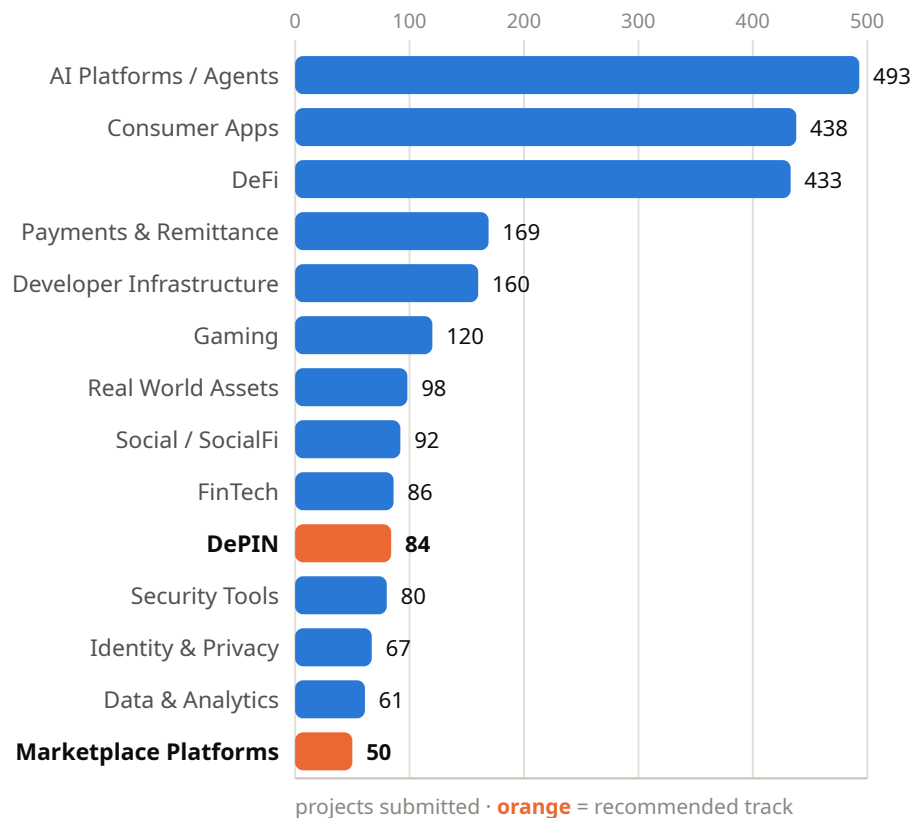


Figure 2. Submissions per track, Frontier hackathon (April 2026, 2,858 projects total). Source: Colosseum Copilot [/filters](#).

Prior art: what has already been tried

READ THIS FIRST

These are hackathon submissions — demos and prototypes, not production products. Many may no longer be active. Most hackathon projects don't turn into successful startups. They are here to show what has been tried and to save you time, not to scare you off.

DroneForce Protocol

`droneforce-protocol:-trustless-coordination-and-on-chain-verification-for-drone-missions` · Breakout, Apr 2025

Your idea, minus agriculture. “Operators discover tasks, complete them, submit flight logs as proof. Smart contracts handle escrow and automated payouts, Arweave stores proof immutably.” Primitives: escrow + oracle. Stack: Solana + Arweave. Team of two. **Did not win.** Watch their technical demo before you write a line of code.

AGRIV Smart Pesticide Spraying Robot

`agriv-smart-pesticide-spraying-robot` · Breakout, Apr 2025

Autonomous ground spraying for Indian agriculture with a DePIN shared-infrastructure layer. Tagged [hardware-oracle](#) and [token-incentives](#). Did not win. Same problem, ground robots instead of drones.

Zeta-X

zeta-x · Radar, Sep 2024

Decentralized drone-based food delivery with crypto payments. Different payload, same “pay a drone operator on-chain” shape.

SkyTrade

skytrade-drones-radar-and-air-rights-marketplace · Renaissance, Mar 2024

Real-time drone tracking and tokenized air rights. The only prior project that took the airspace and regulatory layer seriously.

DeGarden · Sparked Sense · SeedFlow · Umay

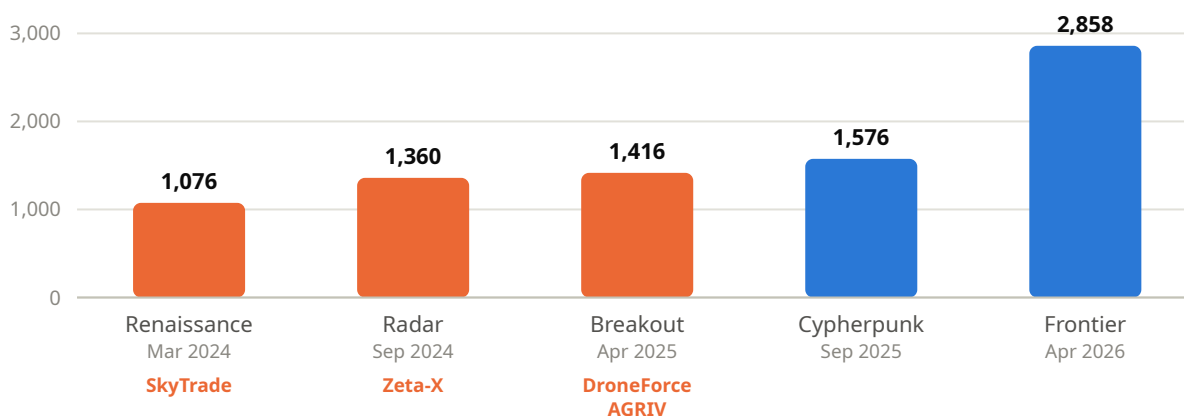
degarden · sparked-sense · seedflow · umay

Soil-sensor DePIN, verifiable IoT data on Solana, smallholder micro-lending with parametric insurance, and tokenized fractional agriculture. Adjacent on the data and financing sides.

SOL CHARGE · Parkchain

sol-charge · parkchain

Asset-utilization marketplaces for EV chargers and parking. Both chose stablecoins over a native token. Worth understanding why.



Editions carrying a prior drone submission are marked in orange. None of those projects placed.

Figure 3. Submissions per hackathon edition. Every prior drone attempt built the coordination layer and treated verification as “upload the flight log.” None placed. The verification handwave is the recurring failure mode — and it is the same handwave in your submission as written.

Accelerator check

No drone or agriculture-services overlap in the Colosseum accelerator portfolio; eleven results, all tangential. The adjacent companies are **DeCharge** ([decharge](#) , decentralized EV charging, now a Grid-indexed live product with two hardware SKUs), **DeVolt** ([devolt](#) , energy trading) and **CarbonPay** ([carbonpay](#)). All three are “coordinate independently-owned physical assets and settle on Solana” — the same structural problem with a different asset.

STUDY THIS ONE

DeCharge is the single best company for you to study. It solved hardware distribution and operator onboarding in an emerging market, which is exactly your hard part.

What the archives say

The verification problem is formally unsolvable as you have specified it

a16z, “Manipulated signals in DePIN protocols” (Milionis, Ernstberger, Bonneau, Kominers, Roughgarden, June 2025) is the single most important paper for your idea. It proves that truthful elicitation is unsolvable unless the setting satisfies *source identifiability*: the source must not be able to manipulate signals so that two different true states look identical to all observers.

THE FATAL FLAW, STATED PRECISELY

A drone flying the spray pattern with the pump off produces a flight log identical to a drone that sprayed. Under source identifiability, **no mechanism — no token, no slashing, no oracle committee — can elicit the truth from that signal set**. This is not a gap in your engineering. It is a property of the signal you chose.

Peer-to-peer is why you don’t need a token — and why one would hurt you

a16z, “Why DePIN matters, and how to make it work” (Guy Wuollet, March 2025) draws the distinction that decides your token question. In peer-to-pool, the network must adjudicate disputes, which demands decentralized verification. In peer-to-peer, “the client pays the provider directly ... either party can detect the other party’s lie without needing to prove the lie to the network.”

Your design is peer-to-peer. That is a feature: it means you can ship without solving verification perfectly. But the same source is blunt about the token: peer-to-pool exists largely so a client can pay X while the provider receives $Y > X$, subsidised by speculation. “A peer-to-peer model makes token rewards as a subsidy much harder. If a client can pay X and a provider receives Y , where $X < Y$, and the client and provider can interact directly, then providers will pretend to buy services from themselves.”

WEAKNESS 1 — THE TOKEN

As written, a farmer-operator pair (or one person with two wallets) sprays an empty field and farms the subsidy. Staking and slashing don’t fix this; they set a price for the fraud.

Fix. Settle in USDC. Replace the token with (a) a refundable **performance bond in USDC** that a fraudulent operator forfeits *to the farmer they defrauded* — not to a treasury, which removes the self-dealing incentive entirely — and (b) a **non-transferable reputation record** via Solana Attestation Service, accruing per verified job.

Then say it out loud in the pitch: “We don’t have a token. In a peer-to-peer marketplace a token subsidy is a self-dealing attack surface, so we settle in USDC and put the operator’s own capital at risk instead.” Judges see hundreds of gratuitous tokens. Deliberately not having one, with a rigorous reason, is a differentiator.

The distributor is the go-to-market, and the archives have a name for it

The same paper describes the “**DePIN mullet**”: a centralized company resells the service in fiat while a decentralized network runs underneath, because “most traditional companies have balked at running blockchain full nodes ... They usually don’t understand what crypto is.” Your DJI distributor *is* the mullet. Lead with that.

Szabo’s “**The God Protocols**” (Nakamoto Institute) supplies the trusted-third-party framing that escrow minimization descends from — useful for one grounding line in the pitch, not load-bearing for the design.

Weakness 2: how to actually prove the drone sprayed

Since no single signal can be made unmanipulable, stop trying. Bound the loss instead with a bundle of four signals, none of which is sufficient alone.

- 1. **Flow-meter volume.** Liters actually dispensed, from the spray system’s flow sensor, cross-checked against tank weight delta. This is the signal that distinguishes spraying from flying. **Verify exportability from the Agras in week one.**
- 2. **Volume-over-area consistency.** Dispensed liters ÷ swath-covered hectares must fall inside the agronomic band for the declared chemical and rate. A drone flying dry fails instantly; so does one dumping the tank in a corner.
- 3. **Farmer attestation.** A one-tap confirm in the app. Your strongest and cheapest signal: in a peer-to-peer market the client has both the information and the incentive.
- 4. **Randomized spot check.** For roughly 10% of jobs, the farmer photographs water-sensitive paper cards placed pre-flight; the app scores blue-dot coverage. This is the real agronomic industry standard and it grounds the whole story in something a judge can watch working.

	GPS flight log	Flow volume	Volume ÷ area	Farmer confirm	Spot check
Flies the pattern with the pump off	○ no	● yes	● yes	● yes	● yes
Sprays water instead of chemical	○ no	○ no	○ no	▲ partial	▲ partial
Sprays only half the field	▲ partial	○ no	● yes	● yes	● yes
Sprays a different, cheaper field	● yes	○ no	▲ partial	● yes	▲ partial
Colludes with the farmer on a fake job	○ no	○ no	○ no	○ no	▲ partial

● detected ▲ partially detected ○ not detected

No single signal is sufficient. The bundle is what makes fraud unprofitable rather than impossible.

Figure 4. Which signals catch which fraud modes. Note the bottom row: farmer-operator collusion defeats almost everything, which is precisely why the bond pays out to the counterparty rather than to a treasury — there is then nothing to collude *for*.

HOW TO FRAME IT FOR JUDGES

“We can’t make fraud impossible — that’s a proven impossibility for remote physical work. We make it *unprofitable*: the bond exceeds the job value, the farmer holds the release key, and a dispute freezes the operator’s entire pipeline.” Cited to the paper, this lands far better with sophisticated judges than any claim of trustlessness.

Two more weaknesses worth fixing

Weakness 3 — autonomous docking stations. Drop it, or reduce it to one closing sentence. It signals you’d rather build the fun future than the boring present, and Colosseum explicitly flags “relying on

buzzwords instead of articulating a clear product hypothesis.”

Weakness 4 — chemical liability, which your brief doesn’t mention at all. If a drone sprays the wrong chemical or drifts onto a neighbour’s organic plot, who pays? The moment you escrow the payment you look like a party to the transaction. Realistic v1: you are a coordination layer, operators carry their own liability, you verify but do not warrant. Say it before a judge asks.

The market and the money

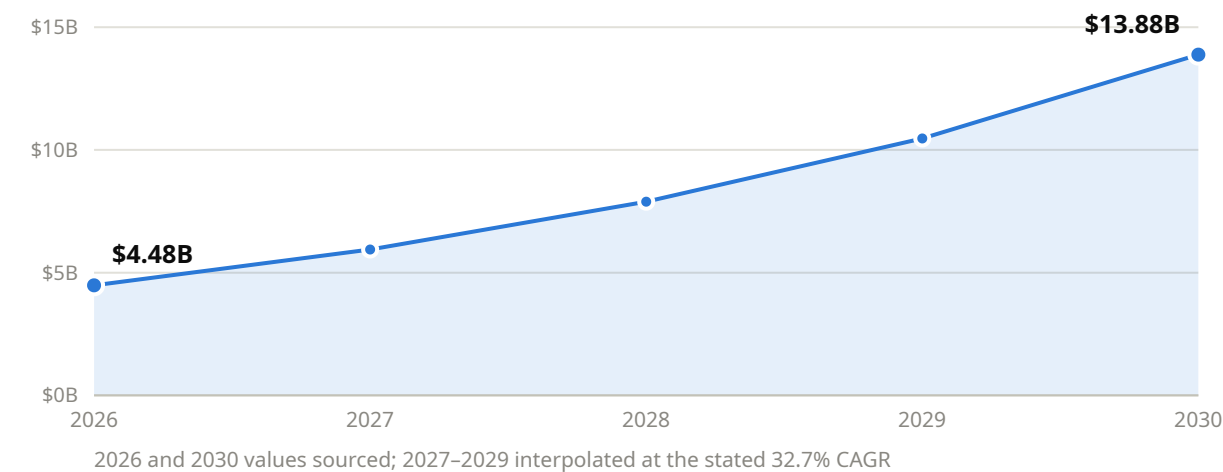


Figure 5. Crop-spraying drone market, 2026–2030. Sources: The Business Research Company; MarketsandMarkets. The drone-services segment — spraying sold as a service rather than hardware sold to owners — is explicitly one of the fastest-growing commercial drone categories.

The operator’s arithmetic is your pitch

This is the part to put on a slide. Every T50 your distributor sells is a \$25,000 asset that needs job flow to pay itself off — which makes you a **sales tool** for the distributor, not a supplicant.

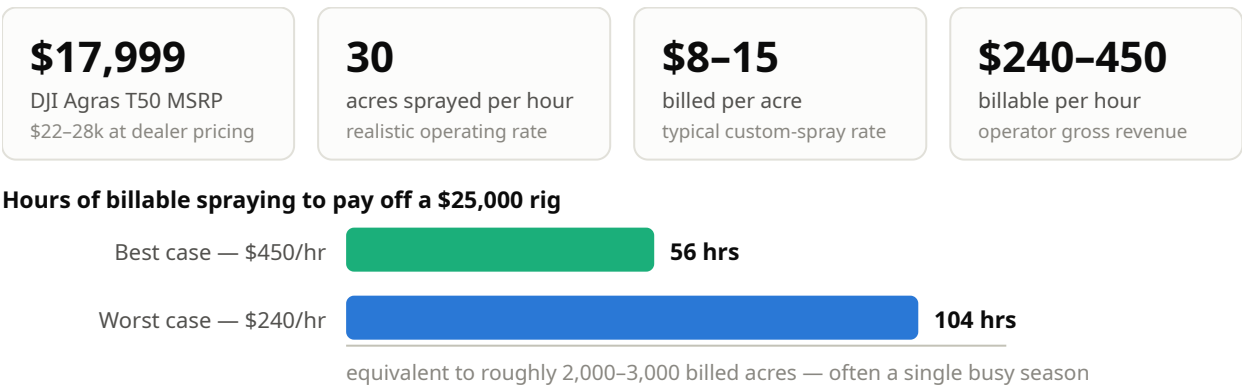


Figure 6. DJI Agras T50 economics at US contract-applicator rates. Georgian rates will be lower; the structure of the argument does not change. The T70P combo runs \$39,999 and covers up to 18 ha/hour.

Revenue model

- **Marketplace fee.** 3–5% of job value at escrow release, in USDC. At Georgian rates of roughly \$10–20/ha on 20 ha jobs, that is \$200–400 per job and \$6–20 to the network.

- **Be honest that this is thin**, then show the second line: a **verified-application-record** subscription for exporters, insurers and cooperatives who need chemical-application compliance data, priced per hectare-season. That is the higher-margin business; the marketplace is the data-acquisition channel.
- **TAM math.** \$4.48B (2026) × ~30% services share × 3% take ≈ **\$40M SAM** on the marketplace fee alone at global full penetration. The record layer is where a larger number lives.
- **Comparables.** Uber (~25% take, too high here), Upwork (~10%), DeCharge (per-session on Solana), AcreConnect (compliance SaaS, no payments).

Landscape classification: differentiation opportunity — geographic and integration

The verification and coordination pieces exist separately in US compliance SaaS and in Solana DePIN tooling. Nothing combines them, and nothing serves the Caucasus and Central Asian smallholder market where spray services are informal and cash-settled.

68

DePIN products tagged Solana on The Grid

43

distinct roots behind them

0

in agricultural field services

2

"drone" hits across 6,300+ Grid products, both irrelevant

BOUNDED CLAIM

"Nobody has built this" here means nobody among 5,400+ Colosseum submissions or 6,300+ Grid-indexed products — not nobody anywhere. Absence of evidence is not evidence of absence, and the pitch should say "as far as we can tell" rather than "no one."

Go-to-market, and the one risk that could sink it

Cold start: supply first, through your distributor

This is the strongest part of your idea and you are underselling it. The distributor isn't just a partner — they are the reason operators exist at all. You are selling them a sales tool: "buy this drone, and here is the job pipeline that pays it off."

Be the demand side yourself for the first season. Sign three to five farms directly, route jobs manually, use the app to settle. Ten real jobs with real settlement beats a polished simulation, and Colosseum explicitly rewards "evidence that a team is solving a real problem for real users."

Network effects here are local, not global. Density in Kakheti doesn't help you in Kenya. Plan a market-by-market rollout and say so; judges respect that more than a world-domination slide.

Regulatory risk — the highest risk, and a current hole in the plan

DO THIS IN WEEK ONE

The GCAA operates the EU-style Open / Specific / Certified framework. The **Open category explicitly excludes any operation that “drops material,”** which puts agricultural spraying categorically outside it. You need the **Specific category**: either a declaration against a published standard scenario, or an operational authorization backed by a submitted risk assessment. Add the weight ceiling — Open tops out at 25 kg and a loaded Agras T50 is far above it — and every operator on your network needs commercial certification plus registration.

Email UAS-GA@gcaa.ge. A screenshot of a reply from the aviation authority in your deck is a serious credibility signal that almost no hackathon team will have. Confirm the pesticide side separately with the National Food Agency — chemical application licensing is distinct from aviation.

RISK	SEVERITY	ASSESSMENT
Regulatory	Highest	Spraying is outside the Open category. Specific-category authorization plus per-operator commercial certification is required before a single legal job runs.
Execution	High	Getting trustworthy telemetry out of a DJI Agras. The flight-record ecosystem is closed. Find this out in week one, not week four — if flow data isn’t exportable, the whole proof story changes.
Technical	Medium	Coordination is easy. Verification is formally hard. Bound the risk rather than claiming to eliminate it.
Market	Medium	Painkiller for the operator with an idle \$25k asset; closer to a vitamin for the farmer, who already has a phone number that works. Anchor the pitch on the operator.

Founder-market fit

What you bring: the distributor relationship. Most crypto founders attacking DePIN have no channel; you have one. **What to close:** you are a front-end developer new to Solana and Anchor, building a system whose core is a Rust program plus hardware telemetry parsing. Get a Rust/Anchor cofounder, or scope the on-chain surface small enough that you can own it. Colosseum’s own guidance: the average winning team size is above three, and solo founders must explain why they are uniquely suited.

The red flag to avoid being: the founder who has never watched a spray job happen. Go stand in a field with an operator before you submit. One photo of you next to an Agras is worth more credibility than any slide.

MVP scope for five weeks

On-chain — an Anchor program of roughly 400 lines

Four instructions. That is the whole surface.

```
JobEscrow (PDA: ["job", farmer, job_id])
├─ fields: farmer, operator, usdc_vault, amount, field_hash,
│         chemical_code, target_rate_l_per_ha, area_ha,
│         proof_hash, state, created_at, dispute_deadline
└─ state: Posted → Accepted → ProofSubmitted → Released | Disputed → Refunded
```

```
Operator (PDA: ["operator", authority])
└─ fields: authority, bond_vault, bond_amount, jobs_completed,
         jobs_disputed, active_job
```

Instructions:

1. `post_job(amount, field_hash, chemical_code, rate, area)`
// farmer funds the USDC vault
2. `accept_job()`
// requires bond $\geq$ job amount; locks the operator
3. `submit_proof(proof_hash, liters_dispensed, area_covered_ha, verdict_sig)`
// program checks liters/area against target_rate tolerance
4. `release()` // farmer confirms → operator paid, bond unlocked
`dispute()` // within window → refund, bond slashed to the farmer
`auto_release()` // permissionless crank after 72h of farmer silence

Deliberately not on-chain: token mint, governance, reputation scoring math, oracle committee, dispute arbitration. The `jobs_completed` / `jobs_disputed` counters on the Operator PDA give you verifiable reputation for free.

Off-chain

- **Farmer and operator app** — Angular PWA. Your Angular/TypeScript background is a real advantage; use it rather than fighting React. Map-based field drawing (Leaflet + turf.js for area), job posting, proof review, one-tap release.
- **Wallet** — embedded or social login, not “install Phantom.” Your users are farmers. This alone will differentiate your demo from most submissions.
- **Proof ingestion service** — Node/TS. Ingests the flight record and flow data, computes liters/ha and coverage, bundles a JSON manifest, pins it to Arweave, hashes it, signs the verdict, returns `(proof_hash, stats, signature)`.
- **Verifier key** — a single service keypair for v1. In the technical demo, say so explicitly and give the path to a committee. Judges reward knowing where your trust assumptions are far more than pretending you have none.
- **Spot-check scoring** — water-sensitive-paper photo to blue-coverage percentage. OpenCV or a plain canvas threshold. Small, visual, and it makes the demo feel real.

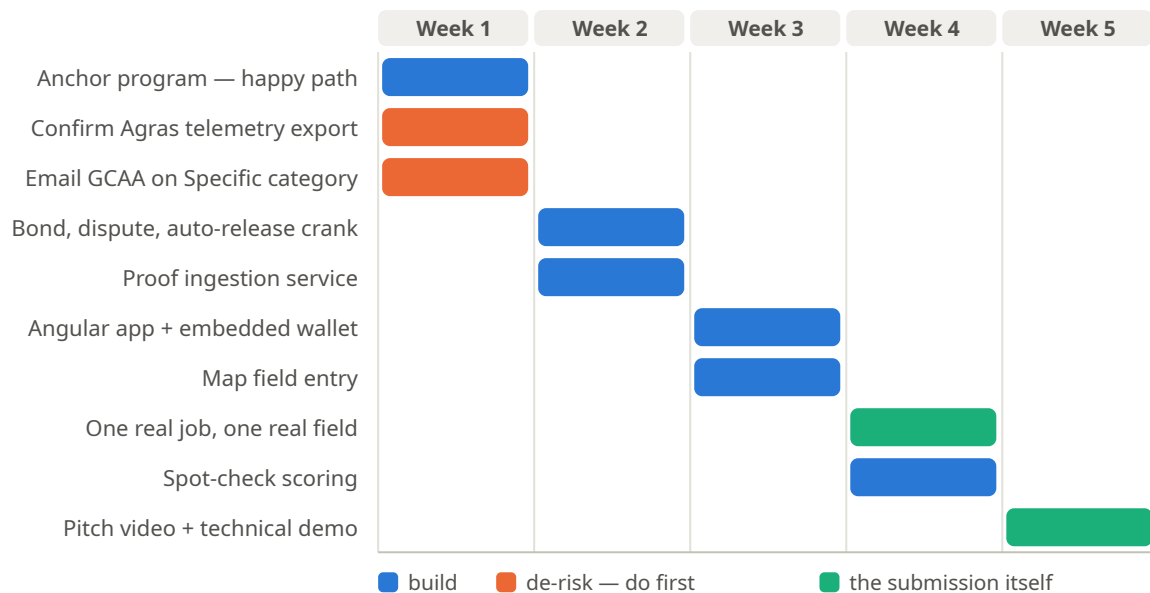


Figure 7. The five-week plan. Week 4 is the whole submission: a single real spray job, settled on devnet and filmed, outweighs every feature you could add instead. Week 5 adds no features.

Stack

Anchor 0.31+ / Rust · TypeScript with [@coral-xyz/anchor](https://github.com/coral-xyz/anchor) · Angular 18 PWA · embedded wallet (Privy/Para, or Solana Mobile if you have a device) · SPL USDC on devnet · Arweave via Irys · Helius RPC and webhooks for state transitions · Leaflet + turf.js · Postgres for off-chain indexing.

What to reuse rather than build

- **Solana Attestation Service** — built for exactly your case: device and location attestations backing DePIN applications with verifiable off-chain proofs. Use it for operator credentials and the per-job verified record instead of rolling your own. Naming it in the technical demo shows you know the current ecosystem.
- **Solana DePIN Quickstart** — the canonical device-signs → aggregator-validates → chain-records pattern. Follow it; don't invent.
- **Switchboard** for oracle plumbing; **Helius** webhooks for reacting to job state changes.
- **DroneForce's repo and technical demo** — same architecture, one hackathon ago. It will save you a week and show you exactly what a judge already saw and passed on.

Pitch video — 3:00 maximum

The pitch video is the most important element of the submission. It is usually the first thing judges review and determines whether you are shortlisted. Treat it as a startup pitch, not a product demo.

0:00–0:20

Who.

"I'm Khristo, in Tbilisi. I work with Georgia's DJI agricultural distributor. Every T50 they sell is a \$25,000 asset that needs \$25,000 of spray jobs to pay for itself — and no reliable way to find them." Open on the relationship. It is your moat.

0:20–0:50

Problem.

A farmer with 20 ha needs four spray passes in a three-day weather window. Finds an operator by phone, pays cash, can't verify anything. The operator can't prove quality, so competes only on price. A lemons market.

0:50–1:30

Solution, with demo footage.

Farmer draws the field, posts the job, funds USDC escrow. Operator accepts, posting a bond. Sprays. The app ingests flow-meter volume and coverage, checks liters/ha, pins the record, submits the hash. Farmer confirms → instant payout. Use the real field footage here.

1:30–2:00

Why crypto — the honest version.

"You could build the payments with Stripe. You couldn't build the *record*. An EU hazelnut buyer needs to trust a chemical-application record without trusting me. That's a verifiable credential, and that's why it's on Solana — sub-cent settlement per hectare, and attestations as native infrastructure."

2:00–2:20

Why now, and traction.

\$4.48B spraying market growing 33% a year. Then: "We ran N real jobs on N real farms this month, settled in USDC." Whatever N is, say it. Add the GCAA reply if you have it.

2:20–2:45

Why us.

Distributor channel, Master's in blockchain, in-market, farmers on speed dial. Introduce cofounders.

2:45–3:00

Roadmap.

Kakheti this season → all of Georgia → Caucasus and Central Asia. One clause on autonomous docking, then stop.

NAMED FAILURE MODES TO AVOID

Exceeding the three-minute limit · flashy visuals with little substance · over-relying on buzzwords · vague or overly technical descriptions · omitting team background · failing to explain the Solana integration clearly · forgetting to grant judges access to your repo, docs and videos.

Technical demo — 2:00

A separate video with a separate job: this one is *how*, not *why*. Do not pitch in it.

0:00–0:15

Architecture.

Diagram on screen. "Four Anchor instructions, one PDA per job, one per operator. Everything else is off-chain by design — here's why."

0:15–0:45

The state machine, in the code.

Posted → Accepted → ProofSubmitted → Released, with the dispute and auto-release branches. Show `accept_job` rejecting an operator whose bond is below the job value.

0:45–1:15

The proof pipeline.

Open a real flight record. Show liters dispensed, area covered, the liters/ha computation and the tolerance check inside the program. “A drone that flies the pattern with the pump off produces the same GPS track — that’s why we verify dispensed volume, not flight path.”

1:15–1:40

Settlement on devnet.

Submit the proof. Show the transaction in the explorer, the stored `proof_hash`, the Arweave manifest resolving. Farmer taps confirm; show the USDC transfer and the bond unlocking in the same view.

1:40–2:00

Trust boundaries, said plainly.

“The verifier is one keypair today. In a peer-to-peer market the farmer is the observer with skin in the game, so we don’t need a committee yet. Here’s the path to one.” Close on the impossibility result in a single sentence — it shows you read the literature and chose your design deliberately.

Produced with Colosseum Copilot v1.2.1 on 21 September 2026. Evidence: 5,400+ Colosseum builder-project submissions across five hackathon editions (Renaissance, Radar, Breakout, Cypherpunk, Frontier); The Grid ecosystem index (6,300+ products); the Colosseum crypto archives (a16z crypto research, Nakamoto Institute, Colosseum blog); and live web search. Market figures are third-party projections, not measurements. Hackathon projects cited may no longer be active — verify current status before drawing conclusions.